

Semantic Frame Detection in Literary Text: Identifying Revenge, Hunting, and Travel in *Moby Dick*

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Abstract: A large part of natural language processing (NLP) involves first tokenizing the text, then breaking it down into components, grouping related elements, and finally using statistical methods to train a large language model (LLM) capable of predicting text based on context and other inputs. These models have a wide range of applications, from commercial assistants to data analysis and content generation. In this project, NLP is used to extract and correct semantic information from *Moby Dick* by Herman Melville by identifying occurrences of three specific semantic frames within the text. A custom gazetteer is implemented to correct common misclassifications, such as labeling “Moby Dick” as an organization instead of an animal. Named entities are then organized using the `ne_chunk` function, which clusters them by type. To identify semantic frames, both WordNet and FrameNet are used. WordNet helps group words into synsets (e.g., hypernyms), while FrameNet assigns roles (frame elements) to lexical units, giving context to each frame. This information supports the LLM in understanding the deeper meaning of the text and potentially generating predictive content. The three primary frames analyzed (Revenge, Hunting, and Travel) were selected for their thematic importance and frequency in the narrative. These frames are identified through lexical units, frame elements, and named entity recognition. This approach enables a deeper understanding of the narrative structure and character motivations in literary texts. In this paper, we will explore the design choices, methodological approach, and reasoning behind the frame element (FE) identification criteria developed for each of these three semantic frames.

I. Introduction

Natural Language Processing is a key area of artificial intelligence that enables machines to process and analyze human language in a meaningful way. Modern NLP techniques often combine rule-based linguistic analysis with machine learning algorithms to extract patterns, interpret context, and support applications such as language translation, sentiment analysis, and text classification. A main drawback of this, however, is AI’s lack of understanding when working with predictive text, as the model is run solely on statistics. This reinforces the need to feed context into the model, especially when it is trained on a newer dataset, so accurate results can be produced from the predictive text.

Python has become the dominant programming language for NLP research and development due to its extensive ecosystem of

open-source libraries. Among these, the Natural Language Toolkit (NLTK) is widely used for its comprehensive suite of functions supporting tokenization, part-of-speech tagging, syntactic parsing, and named entity recognition. Built on Python, NLTK provides accessible tools for both novice and advanced users and serves as an effective platform for experimentation in computational linguistics. This paper utilizes NLTK and other Python-based semantic resources to perform semantic frame detection in literary texts, specifically analyzing Herman Melville’s *Moby Dick* through the frames of Revenge, Hunting, and Travel.

II. Background

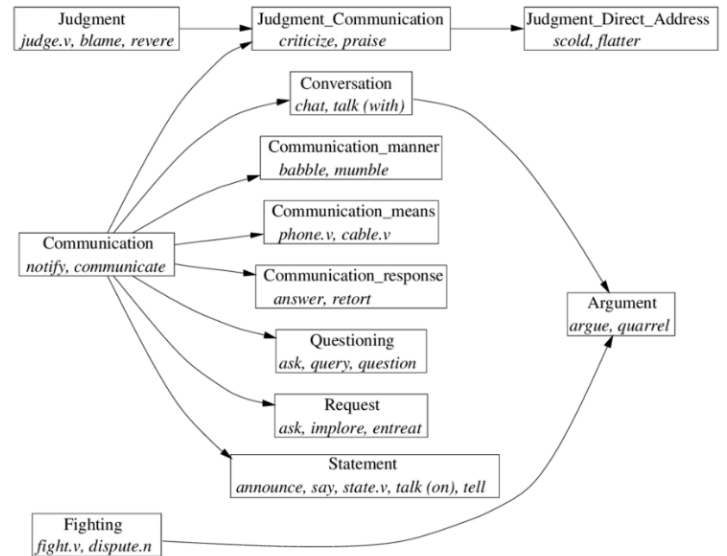


Fig. 1. Semantic network diagram showing relationships among FrameNet communication frames, including judgment, conversation, questioning, and argument categories with their associated verbs.

1.1 Frame Selection and Lexical Units

Frame selection is crucial when analyzing literary texts, as it allows for a more accurate interpretation of narrative elements within their proper context. Understanding the setting, characters, and overarching themes of a work is essential for building reliable and meaningful models. *Moby Dick* tells the story of Ishmael, who joins the whaling ship *Pequod* and becomes caught in Captain Ahab's

obsessive quest for revenge against the great white whale, Moby Dick, who had previously maimed him.

This narrative clearly supports the selection of the semantic frames Revenge, Hunting, and Travel, as they align closely with the central themes of the novel. The Revenge frame is exemplified through Captain Ahab’s relentless pursuit of Moby Dick, driven by a personal vendetta after the whale took his leg. Ahab acts as the avenger, with Moby Dick positioned as the offender, creating a clear instance of this semantic structure. The Hunting frame is equally prominent as the crew of the *Pequod* engages in the act of tracking, chasing, and attempting to kill whales throughout the journey. This includes preparation, pursuit sequences, and detailed encounters with different animal life that reinforce the hunting context. Lastly, the Travel frame appears consistently in Ishmael’s movement from land to sea, his journey aboard the *Pequod*, and his exposure to various ports, cultures, and maritime experiences.

To gather words relevant to the chosen semantic frames of Revenge, Hunting, and Travel, it was necessary to collect lexical units, which are the specific words that evoke a particular frame, such as “hunt,” “pursue,” or “track” for the Hunting frame. This process was conducted using FrameNet, which organizes vocabulary into frames and provides a list of lexical units associated with each frame. By using the .lexUnit attribute, specific words tied to each frame could be accessed and compiled for analysis. WordNet, on the other hand, organizes words into synsets (sets of synonyms) and provides semantic relationships such as hypernyms and hyponyms. It was used to supplement the vocabulary list with related terms using a command like `wn.synsets('travel')`.

1.2 Frame Element Identification

Frame Element identification was implemented through an approach tailored to the specific requirements of each semantic frame. The methodology utilized Part-of-Speech (POS) tagging combined with WordNet semantic relationships to identify core frame elements, particularly for noun-based elements such as "prey" in Hunting frames, where WordNet's hyponym and meronym relationships found the distinctions between different types of closely related scenarios. Named Entity Recognition (NER) captured person entities such as protagonists and antagonists in Revenge frames.

Both lemmatization and stemming techniques were utilized for lexical normalization, where lemmatization reduces words to their dictionary form (e.g., "running" to "run") while stemming removes suffixes to find root forms (e.g., "running" to "runn"). Lemmatization preserves semantic meaning and produces valid words, whereas stemming is faster but may create non-words. Many instances of semantic frame relations were observed, such as the inheritance, or “is-a,” relation in which one frame functions as a subtype of another. For example, the Revenge frame in *Moby-Dick* inherits from the broader Rewards and Punishments frame, linking specific acts of vengeance to a more general framework of consequences and retribution.

Regular expressions were explored to create frame-specific matching rules for complex multi-word expressions that could indicate frame elements across various frames. This strategy complements the frames() function, as using a specific regex pattern can find related frames that match the pattern. Proper text parsing was essential to handle sentence boundaries and tokenization, ensuring accurate identification of frame boundaries and their constituent elements within the corpus.

NLP Techniques for Frame Element Identification

#	Technique	Definition	Role in FE Identification
1	Text Normalization	Standardizes text format (case, punctuation, encoding)	Prepares raw text and removes processing noise
2	Text Segmentation	Breaks text into words and sentences	Creates discrete units for frame analysis
3	Lemmatization/Stemming	Reduces words to root forms	Normalizes variations for better term matching
4	POS Tagging	Assigns grammatical categories to words	Identifies syntactic roles for frame elements
5	NER	Identifies named entities (persons, locations, etc.)	Captures frame elements like protagonists, places
6	WordNet	Uses semantic word relationships	Expands matching beyond exact terms
7	Regular Expressions	Matches complex text patterns	Captures frame-specific linguistic patterns
8	FrameNet	Applies semantic frame theory	Provides framework for frame element categories
9	Parsing (CFG)	Analyzes sentence structure	Determines syntactic boundaries and relationships

Fig. 2. Nine NLP techniques for Frame Element identification with their definitions and roles in semantic frame analysis.

1.3 Overall System Design

The process of developing the small language model began by inputting the literary text, which ultimately resulted in sentences that only correlated with the three specific semantic frames. The corpus was inputted as a text file and then segmented using sentence tokenization, followed by word tokenization to facilitate linguistic processing. To allow for many of the steps, several Python libraries were imported, including requests for retrieving web-based text data, BeautifulSoup for HTML parsing, nltk modules such as `sent_tokenize`, `word_tokenize`, `pos_tag`, and `ne_chunk` for text preprocessing and named entity recognition, Tree for syntactic structure parsing, and WordNet for lexical database access and synonym retrieval.

The semantic analysis focused on three frames from FrameNet, Revenge, Hunting, and Travel, chosen for their thematic relevance to the narrative. Preprocessing involved the application of part-of-speech tagging to each tokenized sentence using `pos_tag()`, which assigns grammatical categories to individual words and aids in lexical disambiguation. Named Entity Recognition (NER) was then performed using `ne_chunk()`, which constructs hierarchical parse trees for detecting entities such as persons, locations, and organizations. Following initial NER output, an evaluation of the results revealed consistent misclassifications; for instance, entities like "Kraken" were mistakenly labeled as organizations rather than animals. To address this, a custom gazetteer was developed that mapped misidentified entities to corrected labels, enabling more accurate semantic frame detection.

After the gazetteer was plugged into the code, the final step was compiling the lexical units into a list; each sentence in the corpus was iterated over using a counter variable to assign a unique sentence number. The system then checked whether the sentence contained an occurrence of any of the three target semantic frames and, if so, displayed only those sentences. For each detected frame occurrence, the output included the sentence number, the name of the frame, the lexical unit that triggered it, and the corresponding frame elements

paired with the specific words or named entities that fulfilled each role.

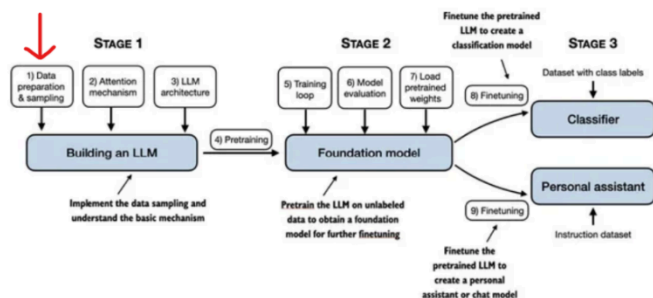


Fig. 3. Three-stage LLM development pipeline from initial building through foundation model creation to specialized applications via finetuning. The red arrow marks where the experiment of this paper takes place.

III. Results and Observations

The detection of semantic frames was applied to the entirety of *Moby Dick* to identify instances of the frames through lexical unit matching and contextual analysis, first by being broken down into smaller text chunks and then being done as a whole. The initial stage of analysis involved tokenizing the entire text of *Moby Dick* and applying syntactic chunking to extract noun and verb phrases. This allowed for a detailed breakdown of each sentence into discrete grammatical components, facilitating both named entity recognition (NER) and later semantic role labeling. However, early trials revealed that many specific terms, especially proper nouns central to the novel, were frequently misrepresented, usually being placed in the “organization” chunk. For instance, the default NLP pipeline often mislabeled “Leviathan” as an organization, “Pequod” as a corporate entity, and sometimes failed to identify “Ahab” as a person. These misclassifications had cascading effects on semantic frame parsing, leading to incorrect and incomplete frame element assignments.

The main reason this was a major problem was when detecting frames: Ahab might be excluded from the *Avenger* role in a *Revenge* frame, or the *Vehicle* element in a *Travel* frame might remain empty if “Pequod” was not recognized as a ship. To address these shortcomings, a custom gazetteer was developed, tagging key entities, such as characters, ships, places, and whales, with appropriate labels, effectively overriding erroneous default classifications. Once implemented, the gazetteer significantly improved the accuracy of both NER and frame element identification, allowing the system to better capture relationships among characters and events within the novel. Following these corrections, the system was able to extract a substantial number of meaningful semantic frames. Among these, the *Hunting* frame appeared most frequently, which is consistent with the novel’s dominant themes of pursuit and obsession. In contrast, the *Revenge* frame was less common but more concentrated, often clustered around peaks involving Captain Ahab.

An additional pattern emerged regarding the complexity of the source material. Sentences with straightforward grammatical structure

yielded more accurate frame identification, whereas sentences featuring poetic constructions, embedded clauses, or figurative language posed challenges for lexical unit disambiguation. Notably, frame detection was often more robust when explicit action verbs were used, as opposed to metaphorical or implied references, as this made it harder for the model to understand.

Despite these challenges, the analysis revealed several interesting patterns. The *Revenge* frame, while rarer in frequency, was often expressed with greater lexical variety than initially expected, suggesting a deeper semantic field. The *Travel* frame frequently overlapped with the *Hunting* frame, indicating that frame co-occurrence could be a meaningful dimension to explore in future work. These findings highlight both the strengths and limitations of the current rule-based system. While lexical unit matching provides a strong foundation for frame detection, its effectiveness is constrained by the accuracy of named entity recognition, the expressiveness/content of the gazetteer, and the actual understanding behind what the text is relaying.

VI. Conclusion

This project successfully demonstrated automated semantic frame detection within literary texts using a combination of NLP techniques and lexical-semantic resources. The implementation identified three semantic frames across a literary corpus by leveraging multiple sources to locate frame-related terms and their associated elements. The custom gazetteer proved crucial for handling literary-specific vocabulary and character names that standard NER systems often misclassify. Large Language Models present transformative potential across domains, as they can learn contextual patterns that manual rules cannot capture. With continued training and fine-tuning, LLMs could automatically detect semantic frames in applications ranging from information extraction and question answering to dialogue systems and content analysis. As these models advance, they will enable scalable semantic understanding across diverse text types, making frame detection accessible for real-world applications that surpass the capabilities of traditional rule-based approaches.

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